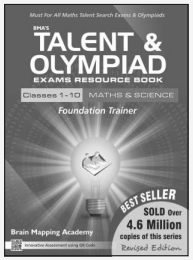
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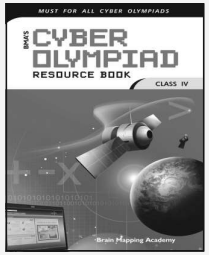
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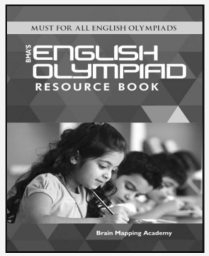
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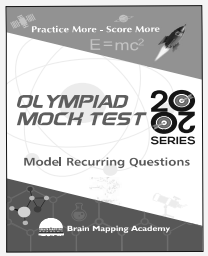
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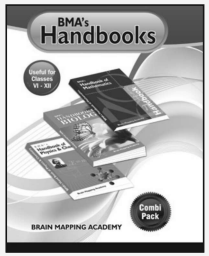
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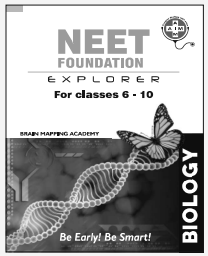
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Class 6

Please fill the following details immediately

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Questions : 50 **Time : 60 minutes**

INSTRUCTIONS

Read all instructions carefully before attempting any question.

- Ensure that the 'Class' printed here and inside, is the same as the test you are appearing for.
- You must complete the paper within the time allotted.
- Do not open this question paper until you are permitted to.
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- Figures herein are not to scale. Hence, you cannot depend on the estimate of size or measurement. Use your knowledge of the subject.
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Class – 6 Mathematics - 1

*Questions from **1 to 30** are single answer questions, darken the correct option as your answer in the OMR sheet provided. Each carries **1 mark**.*

01 The sum of a number and its reciprocal is $\frac{10}{3}$ then select the numbers.

(A) $3, \frac{1}{3}$ (B) $3, \frac{-1}{3}$ (C) $-3, \frac{1}{3}$ (D) $-3, \frac{-1}{3}$

02 A taxi charges ₹ 25 for the first kilometre, and ₹ 18 for each kilometre there after. What would be the taxi charge for a trip of 10 km ?

(A) ₹ 250 (B) ₹ 180

(C) ₹ 187 (D) ₹ 197

03 Select option correctly showing the value of 6.78×10^5

(A) 6,78,00,000 (B) 6,78,000

(C) 0.0000678 (D) 0.000000678

04 What number has to be added to 345670 in order to make it divisible by 6 ?

(A) 2 (B) 4

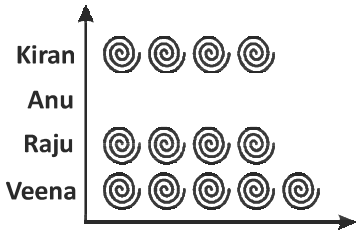
(C) 5 (D) 6

05 A seven digit number 876p37q is divisible by 225. Identify the value of p and q.

(A) 9, 0 (B) 0, 0

(C) 0, 5 (D) 9, 5

06 The given pictograph is incomplete. It shows the number of symbols each child has.



Key : Each ◎ represents 2 sweets.

The total number of sweets the four children have is 30. How many symbols does Anu have ?

(A) 2 (B) 4 (C) 6 (D) 8

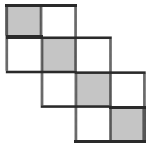
07 Operations are placed in each ◯ so that 3 ◯ 5 ◯ 7 ◯ 9 = 78. Listed from left to right. Identify the operations.

(A) +, × + (B) +, +, ×

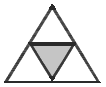
(C) ×, ×, – (D) ×, +, ×

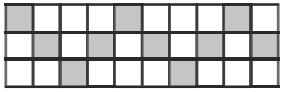
Space for Rough work

08 **Identify the figure which represents 0.4 ?**

(A) 

(B) 

(C) 

(D) 

09 **Identify the product of $(3xy)$ and $(5x^2 + 2xy + 3y^2)$**

(A) $(15x^3y + 6x^2y^2 + 9xy^3)$

(B) $(15x^3y + 2xy + 3xy^2)$

(C) $(15x^2y + 6x^2y^2 + 9xy^2)$

(D) $(15x^2 + 5xy + 3y^2)$

10 **The angles of a ΔABC measure 27°, 43° and 110°. What kind of triangle is ΔABC ?**

(A) Equilateral (B) Acute

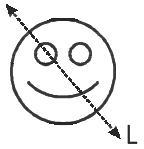
(C) Obtuse (D) Right

Space for Rough work

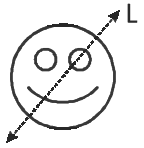
11 **What is the value of the expression 5(a + b) – 3(b + c) if a = 4, b = 3 and c = 2 ?**

(A) 20 (B) 18 (C) 14 (D) 10

12 **Which option is the line L most likely a line of symetry ?**

(A) 

(B) 

(C) 

(D) 

13 **Cherry used a copy machine to enlarge a drawing to 150% of the original size. If the width of the original drawing was 37 centimetres, Identify the width of the copy of drawing.**

(A) 37 cm (B) 55.5 cm

(C) 92.5 cm (D) 23.3 cm

Space for Rough work

14 **Which option has the greatest value ?**

(A) $\frac{5555}{6666}$ (B) $\frac{666}{777}$ (C) $\frac{77}{88}$ (D) $\frac{8}{9}$

15 **Which is the longest rectangle ?**

(A) Area = 24 cm² } 2 cm (B) Area = 48 cm² } 6 cm

(C) Area = 100 cm² } 8 cm (D) Area = 60 cm² } 8 cm

16 **In a class of 49 students, the ratio of girls to boys is 4 : 3. If 4 girls leave the class, choose the ratio of girls to boys.**

(A) 11 : 7 (B) 8 : 7 (C) 6 : 5 (D) 9 : 8

17 **If $x \neq 0$. then find the value of $\frac{2x^4 - 6x^3 + 4x^2 + 10x}{2x}$.**

(A) $(x^3 - 3x^2 + 2x + 5)$ (B) $(x^3 - 6x^3 + 4x^2 + 5x)$

(C) $(2x^3 - 6x^2 + 4x + 5)$ (D) $(2x^4 - 6x^3 + 4x^2 + 5x)$

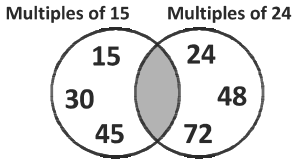
18 **The base of a triangle is 3 cm more than its height (h). Select option correctly shows the area of the given triangle.**

(A) h(h + 3) cm² (B) $\frac{1}{2}$h(h + 3)cm²

(C) h(h – 3)cm² (D) $\frac{1}{2}$h(h – 3)cm²

Space for Rough work

19 **Saroja is filling numbers in Venn diagram. No number is to be entered more then once.**



What is the least number than can be appropriately place in the shaded area of the diagram ?

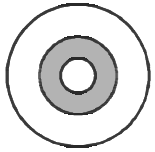
(A) 360 (B) 240 (C) 120 (D) 60

20 **If a : b = 3 : 5 and b : c = 7 : 8 then identify the equivalent ratio of 2a : 3b : 7c.**

(A) 42 : 105 : 320 (B) 15 : 21 : 35

(C) 6 : 15 : 40 (D) 30 : 21 : 350

21 **In the diagram, the area of the second middle ring is 6 times the area of the smallest circle. The area of the unshaded outer ring is 12 times the area of the smallest circle. What is the ratio of the largest circle area and smallest circle area ?**



(A) 3 : 1 (B) 6 : 1 (C) 18 : 1 (D) 19 : 1

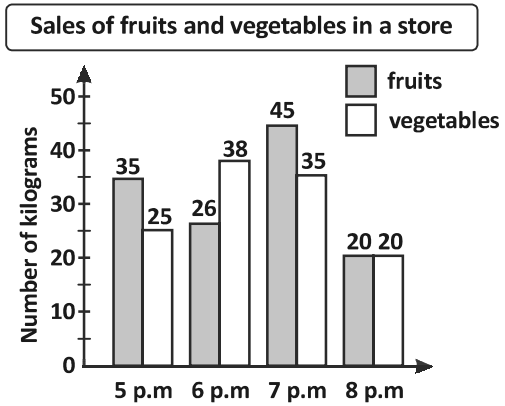
Space for Rough work

22 **A rectangular poster is 1.5 m long and 90 cm wide. Select option correctly showing the ratio (in lowest terms) of the length and perimeter of the poster.**

(A) 1 : 4 (B) 4 : 1

(C) 5 : 16 (D) 7 : 11

23 **How many kilograms of fruits were sold during the four hours ?**



(A) 120 kg (B) 124 kg

(C) 126 kg (D) 144 kg

Space for Rough work

24 **There are six integers whose product is 1. Which of the following cannot be the sum of those six integers ?**

(A) –6 (B) –2 (C) 0 (D) 2

25 **Select the value of (2 + 4 + 6) – (1 + 3 + 5).**

(A) 0 (B) 3

(C) –3 (D) –1

26 **Which integer is closest in value of $\frac{43}{4}$?**

(A) 11 (B) 10

(C) 9 (D) 8

27 **On subtracting 8 from –8, we get**

(A) 0 (B) –16

(C) –64 (D) –1

Space for Rough work

28 If $x : y = 3 : 5$ and $y : z = 5 : 7$ then identify the equivalent ratio of $(y - x) : (x + z)$.

(A) 2 : 23 (B) 1 : 5

(C) 1 : 6 (D) 15 : 46

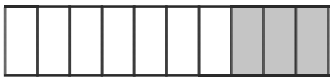
29 The HCF of two expressions p and q is 1. Choose option that correctly showing the LCM of p and q.

(A) p + q (B) p – q (C) pq (D) $\frac{1}{pq}$

30 Which figure shows $\frac{1}{3}$ shaded ?

(A) 

(B) 

(C) 

(D) 

Space for Rough work

Class – 6 **Mathematics - 2**

*Questions from **31** to **35** have more than one correct option, darken the correct options as your answer in the OMR sheet provided. Each carries **3 marks**.*

31 Which operations satisfy closure property of whole numbers ?

(A) Addition (B) Subtraction

(C) Multiplication (D) Division

32 Which options represent an integer ?

(A) $1\frac{1}{2}+\frac{5}{2}$ (B) $\frac{12}{17}-\frac{7}{34}$

(C) $-2-\frac{1}{2}+\frac{3}{2}$ (D) $10-\left(-\frac{12}{3}\right)$

33 Which of the following is an even number ?

(A) Even × odd × even (B) Even × odd × odd

(C) Even × even × even (D) Odd × odd × odd

34 Which options are solids ?

(A) Cube

(B) Cuboid

(C) Cylinder

(D) 6th class Mathematics text book

35 The measure of one angle of an isosceles triangle is 50°. Identify the measure of the other angles in this triangle.

(A) 50° and 90° (B) 50° and 80°

(C) 30° and 80° (D) 65° and 65°

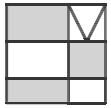
Class – 6 — Reasoning

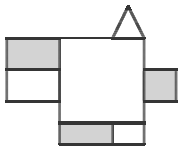
*Questions from **36 to 45** are single answer questions, darken the correct option as your answer in the OMR sheet provided. Each carries **1 mark**.*

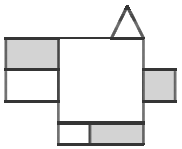
36 **Which is odd one out ?**

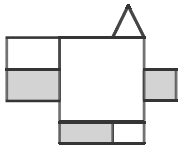
(A) PY (B) IR (C) FG (D) BK

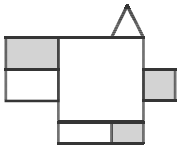
37 **How does the given shape look like when opened out ?**



(A) 

(B) 

(C) 

(D) 

Space for Rough work

38 **Find the mirror image to the given figure.**



(A) 

(B) 

(C) 

(D) 

39 **If 'paper' is called 'wood', 'wood' is called 'straw', 'straw' is called 'grass', 'grass' is called 'rubber' and 'rubber' is called 'cloth', what is furniture made up of**

(A) grass (B) straw

(C) wood (D) paper

40 **Complete the given series.**

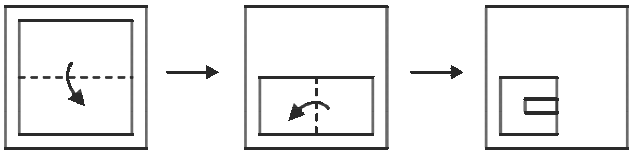
9 11 13 14 17 17 ?

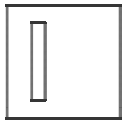
(A) 20 (B) 22

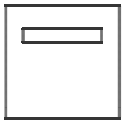
(C) 21 (D) 19

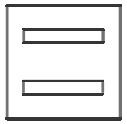
Space for Rough work

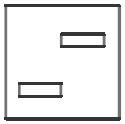
41 A piece of paper is folded and punched as shown below in the question figures. Which option indicate how it will appear when opened ?



(A) 

(B) 

(C) 

(D) 

42 Rearrange the following jumbled letters and choose the middle letter of the correct word.

GMOAN

(A) N (B) G

(C) M (D) A

Space for Rough work

43 If 8 × 4 = 42, 6 × 6 = 33, 2 × 2 = 11, 4 × 6 = 23 then 2 × 8 = ?

(A) 21 (B) 28

(C) 14 (D) 42

44 Which letter is two to the right of the letter immediately to the left of the letter three to the right of the letter C ?

A B C D E F G H

(A) G (B) H

(C) F (D) C

45 Find the matching pair.

21 : 3 :: 574 : ?

(A) 23 (B) 82

(C) 97 (D) 113

Space for Rough work

Class – 6 — Critical Thinking

*Questions from **46 to 50** are single answer questions, darken the correct option as your answer in the OMR sheet provided. Each carries **1 mark**.*

46 **Which shape fit the given incomplete jigsaw ?**



(A) 

(B) 

(C) 

(D) 

47 **In the number sequence, 1, 1, 2, 3, 5, 8, 13 .....**
What is the 14th term ?

(A) 144 (B) 377 (C) 199 (D) 233

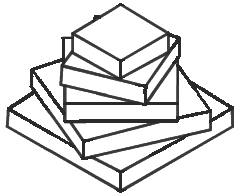
48 **Today is 21st August. The day of the week is Monday. This is a leap year. What will be the day of the week on this day after three years ?**

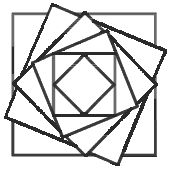
(A) Monday (B) Thursday

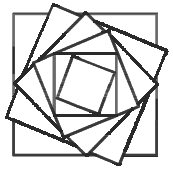
(C) Wednesday (D) Saturday

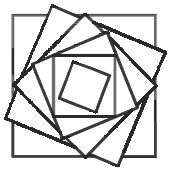
Space for Rough work

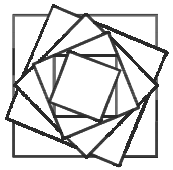
49 **Five blocks are placed one above another in a particular sequence (as shown in figure). If a sixth block is added to the same sequence, which would be the correct top view ?**



(A) 

(B) 

(C) 

(D) 

50 **You are living in a school hostel. The dal served to you in the mess has a lot of stones. What would you do ?**

(A) Leave eating the dal altogether.

(B) Bring the matter to the notice of mess incharge.

(C) Speak to the cook about changing the dal.

(D) Buy your own dal and cook it in your room.

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UNIFIED INTERNATIONAL MATHEMATICS OLYMPIAD (UPDATED)

CLASS - 6

Question Paper Code : UM9009

KEY

1	2	3	4	5	6	7	8	9	10
A	C	B	A	C	A	D	A	A	C
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
A	D	B	D	C	B	A	B	C	C
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
D	C	C	C	B	A	B	B	C	B
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
A,C	A,C,D	A,B,C	A,B,C,D	B,D	C	A	B	B	C
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
C	A	C	A	B	D	B	B	C	B